

TECHNOKRUTI

Algorithms and Pseudocode

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Faculty coordinators:

Event Description :

This event aims to test participants' problem-solving skills by designing and expressing algorithms through pseudocode. Participants will tackle a range of challenges, from simple logical problems to complex real-world scenarios, showcasing their ability to think critically and optimize solutions.

Whether you're a beginner or a seasoned coder, this challenge offers a unique opportunity to sharpen your algorithmic thinking, improve your approach to problem-solving, and learn from peers.

Event Activities :

- **Pseudocode design challenge:** Participants will be given a simple algorithm and will be asked to convert it into pseudocode
- **Design Challenge:** Participants are given a problem statement and asked to design an efficient algorithm to solve it. The problem can range from simple puzzles to complex real-world scenarios.
->Example of challenges : Finding largest number in array, Find missing number in array, Travelling salesman problem.
- **Optimization Round:** Provide participants with a basic algorithm and challenge them to optimize it for time and/or space complexity. This encourages thinking about different ways to improve existing solutions.
->Problems: May be difficult for SE students
- **Presentation and Explanation Round:** Participants present their algorithms and explain the logic and thought process behind them. This can be done individually or in teams.

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Judging Criteria :

- **Correctness** : Does the algorithm solve the problem as intended? It should handle all edge cases effectively and provide the correct output for a variety of inputs.
- **Efficiency** : How efficient is the algorithm in terms of time and space complexity? Judges should consider the Big-O notation and the practicality of the solution for large inputs.
- **Clarity and Readability** : Is the pseudocode or algorithm clearly written and easy to understand? Proper use of variables, clear step-by-step instructions, and good formatting are important.
- **Edge Case Handling** : How the algorithm handles edge cases, such as null values, different data types, large values, etc.
- **Creativity and Innovation** : How creatively does the participant approach the problem? Unique or unconventional solutions that solve the problem effectively should be rewarded.
- **Presentation Skills**: How well does the participant explain their algorithm? Are they able to articulate their thought process and justify their choices?